Unit 4: Electric and Magnetic Circuits

1. Basic Electric Circuit Quantities

Definition: An electric circuit is a closed path through which electric current flows. The fundamental quantities that define this flow are EMF, Current, Potential Difference, Power, and Energy.

- **EMF (Electromotive Force):** The total driving force provided by a source (like a battery or generator) that pushes electrons through a circuit. It is measured in **Volts (V)**.
- **Current (I):** The rate of flow of electric charge (electrons) through a conductor. Measured in **Amperes (A)**. Formula: $I = Q/t$.
- **Potential Difference (V):** The difference in electric potential energy between two specific points in a circuit. It is the work done to move a unit charge between these points. Measured in **Volts (V)**.
- **Power (P):** The rate at which electrical energy is consumed or generated by a circuit component. Measured in **Watts (W)**. Formula: $P = V \times I$ or I^2R.
- **Energy (E):** The total amount of work done by the electrical power over a specific period of time. Measured in **Joules (J)** or **Kilowatt-hours (kWh)**. Formula: $E = P \times t$.

👉 **Hinglish Explanation:** EMF battery ki wo puri taqat hai jo electrons ko push karti hai. Jab electrons behte hain toh use Current kehte hain. Kisi component (jaise bulb) ke dono taraf jo voltage drop hota hai, use Potential Difference kehte hain. Bulb kitni tezi se energy kha raha hai wo Power (Watt) hai, aur kitne ghante tak khayi wo Energy (Unit/kWh) ban jati hai jiska humara bijli ka bill aata hai.

2. Magnetic Circuit Fundamentals

Definition: A magnetic circuit is the closed path followed by magnetic lines of force (magnetic flux). It is very similar in concept to an electric circuit.

Key Magnetic Quantities:

- **M.M.F. (Magnetomotive Force):** The force that drives the magnetic flux through a magnetic circuit. It is equivalent to EMF in an electric circuit.
  Formula: $MMF = N \times I$ (where N = number of turns, I = current). Unit: **Ampere-turns (AT)**.
- **Magnetic Force / Field Intensity (H):** The MMF per unit length of the magnetic circuit.

Formula: $H = \frac{NI}{L}$. Unit: **AT/meter**.

- **Permeability (μ):** The ability of a material to conduct magnetic lines of force. It tells us how easily a material can be magnetized.
  Formula: $\mu = \mu_0 \times \mu_r$ (where μ_0 is absolute permeability of free space, μ_r is relative permeability).
- **Reluctance (S):** The opposition offered by a magnetic circuit to the establishment of magnetic flux. It is equivalent to Resistance in an electric circuit.
  Formula: $S = \frac{L}{\mu A}$ or $S = \frac{MMF}{Flux}$. Unit: **AT/Weber**.
- **Leakage Factor:** In a real magnetic circuit, not all flux follows the intended core path; some leaks into the surrounding air (Leakage Flux).
  Formula: $Leakage Factor (\lambda) = \frac{\text{Total Flux}}{\text{Useful Flux}}$. Its value is always > 1 (usually 1.1 to 1.25).

👉 **Hinglish Explanation:** Jaise electric circuit me EMF current chalata hai, waise hi magnetic circuit me **MMF** magnetic flux chalata hai. MMF turns aur current ko multiply karke nikalte hain. **Reluctance** resistance jaisa hai jo flux ko rokta hai. **Permeability** batati hai ki material kitna achha rasta dega. **Leakage factor** matlab jo flux hawa me waste ho gaya (leak ho gaya) uski calculation.

3. B-H Curve and Hysteresis Loop

Definition: The B-H curve is a graphical representation of the relationship between Magnetic Flux Density (B) and Magnetic Field Intensity (H) for a specific magnetic material.

When a magnetic material (like an iron core) is subjected to an alternating magnetic field, it does not magnetize and demagnetize in a straight line. The flux density (B) lags behind the magnetizing force (H).

Hysteresis Loop Details:

- **Initial Magnetization:** As H increases from zero, B increases non-linearly until it reaches **Saturation** (where increasing H no longer increases B).
- **Retentivity (Residual Magnetism):** When H is reduced back to zero, the material remains slightly magnetized. The remaining value of B is called Retentivity.
- **Coercivity (Coercive Force):** The amount of negative magnetic force (-H) that must be applied to completely wipe out the residual magnetism (make B = 0).

Hysteresis Loss: When AC current is applied, the material goes through this loop continuously. The area *inside* the hysteresis loop represents the energy lost as heat during each cycle. Materials used for transformers (like Silicon Steel) are chosen for having a very "narrow" hysteresis loop to minimize this energy loss.

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B-H CURVE (HYSTERESIS LOOP):

               +B (Flux Density)
                |      Saturation
                |     /|
   Retentivity  |___.´ |
                |  /   |
         _______| /____|_________ +H (Magnetizing Force)
        |       |/
```

```
        |     .´|   <- Coercivity
        |    /  |
        |___.   |
                |
               -B
```

> 👉 **Hinglish Explanation:** Jab hum kisi lohe (iron) ko magnet banate hain (H badhate hain), to uske andar ka magnetic power (B) badhta hai. Par jab hum force zero kar dete hain, to loha thoda sa magnet bana reh jata hai (jise **Retentivity** kehte hain). Us bache hue power ko khatam karne ke liye hume ulti direction me force lagani padti hai (jise **Coercivity** kehte hain). Is poore process me jo energy heat ban kar waste hoti hai, use Hysteresis loss kehte hain. Loop ka area jitna patla hoga, loss utna kam hoga.

4. Electromagnetic Induction: Faraday's and Lenz's Laws

Definition: Electromagnetic Induction (EMI) is the phenomenon of producing an induced electromotive force (EMF) in a closed circuit whenever the magnetic flux linked with it changes over time.

Faraday's Laws of Electromagnetic Induction:

1. **First Law:** Whenever the magnetic flux linked with a circuit or coil changes, an EMF is induced in it. If the circuit is closed, an induced current flows.
2. **Second Law:** The magnitude of the induced EMF is directly proportional to the rate of change of the magnetic flux linkage.

$$e = - N \frac{d\Phi}{dt}$$

Where e is induced EMF, N is number of turns, and $d\Phi/dt$ is the rate of change of flux.

Lenz's Law:

The **negative sign** in Faraday's formula is explained by Lenz's Law. It states that the direction of the induced EMF (and the resulting current) is always such that it **opposes the very cause that produces it**. This is a direct consequence of the Law of Conservation of Energy.

> 👉 **Hinglish Explanation:** Michael Faraday ne bataya ki agar aap magnet aur coil ko ek dusre ke aage-peeche hilaoge, toh flux change hoga aur coil me bijli (EMF) ban jayegi. Jitni tezi se hilaoge, utni zyada bijli banegi. **Lenz's law** kehta hai ki jo bijli banegi, wo us magnet ke hilne ka virodh karegi (oppose karegi). Isliye formula me **minus (-)** sign lagta hai. Generators isi principle par kaam karte hain.

5. Types of Induced EMF and Inductance

Definition: Based on how the change in flux is achieved, induced EMF is classified into Dynamically

and Statically induced EMF. Inductance is the property that opposes changes in current.

1. Dynamically Induced EMF:

Generated when a physical conductor **moves** within a stationary magnetic field, cutting the magnetic lines of force.
Application: DC and AC Generators.
Formula: $e = B \cdot l \cdot v \cdot \sin(\theta)$ (where B = flux density, l = length, v = velocity).

2. Statically Induced EMF:

Generated when the conductor is **stationary**, but the magnetic field around it changes (usually by passing alternating current through a coil).
Application: Transformers. It is further divided into:

- **Self-Inductance (L):** The EMF induced in a coil due to the change of current in the *same* coil. It opposes any change in its own current.
  Formula: $L = \frac{N \Phi}{I}$ or $e = -L \frac{di}{dt}$. Unit: **Henry (H)**.
- **Mutual Inductance (M):** The EMF induced in a secondary coil due to the change of current in a neighboring *primary* coil. This is how transformers transfer power without physical connection.
  Formula: $e_2 = -M \frac{di_1}{dt}$. Unit: **Henry (H)**.

👉 **Hinglish Explanation:** Agar taar ghum raha hai aur magnet ruka hai, toh jo bijli banti hai use **Dynamically induced EMF** kehte hain (Generator me). Agar taar ruka hai aur magnetic field change ho rahi hai (AC current ki wajah se), toh use **Statically induced EMF** kehte hain (Transformer me). Khud ki coil me current change hone se EMF banne ko **Self Inductance** aur padosi coil ki wajah se EMF banne ko **Mutual Inductance** kehte hain.

6. Analogy Between Electric and Magnetic Circuits

Definition: To simplify the understanding of magnetic circuits, we can directly compare their properties to the well-known Ohm's Law and parameters of an electric circuit.

This comparison is extremely important for exams and practical circuit analysis.

Electric Circuit	Magnetic Circuit
Driving Force: Electromotive Force (EMF)	**Driving Force:** Magnetomotive Force (MMF)
Flowing Quantity: Current (I = EMF / R)	**Flowing Quantity:** Flux (Φ = MMF / S)
Opposition: Resistance (R = $\rho L / A$)	**Opposition:** Reluctance (S = $L / \mu A$)
Material Property: Conductivity ($\sigma = 1/\rho$)	**Material Property:** Permeability (μ)
Density: Current Density (J = I / A)	**Density:** Magnetic Flux Density (B = Φ / A)
Path: Current actually flows physically.	**Path:** Flux does not physically flow; it is established.
Energy: Energy is constantly consumed to maintain current.	**Energy:** Energy is needed to establish flux, but not to maintain it.

👉 **Hinglish Explanation:** Electric aur Magnetic circuits bilkul judwa bhai jaise hain. Electric me **EMF** dhakka marta hai, Magnetic me **MMF** dhakka marta hai. Electric me **Current** behta hai, Magnetic me **Flux** setup hota hai. Current ko rokne wala **Resistance** hai, wahi flux ko rokne wala **Reluctance** hai. Ye table exam me 100% poochhi jati hai!

🔁 Quick Revision Notes (Exam Saver)

- **Ohm's Law Analogy:** $I = EMF/R$ (Electric) is identical to $\Phi = MMF/S$ (Magnetic).
- **MMF:** $N \times I$ (Ampere-turns). Drives the flux.
- **Reluctance (S):** Opposes flux. $S = L / (\mu A)$.
- **B-H Curve:** Shows non-linear magnetization. Retentivity is residual magnetism; Coercivity is force needed to remove it. Area = Hysteresis loss.
- **Faraday's Law:** $e = -N (d\Phi/dt)$. A change in flux induces EMF.
- **Lenz's Law:** The minus sign in Faraday's law. Induced current opposes its cause.
- **Types of EMF:** Dynamic = Moving conductor (Generator). Static = Stationary conductor, changing field (Transformer).

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